

Cryptography Assignment-3

1) Multiplication Inverse modulo 26 for
1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, mod 26.

*) Inverse of 1

$$26 = 1(26) + 0$$

$$1(1) = 1$$

$$1^{-1} \rightarrow 1 \pmod{26}$$

$$= 1 //$$

*) Inverse of 3.

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2$$

$$1 = 3 - (26 - 3(8))$$

$$2 = 26 - 3(8)$$

$$1 = 3 - 26 + 3(8)$$

$$1 = 3(9) + 26(-1)$$

$$3^{-1} \rightarrow 9 \pmod{26}$$

$$= 9 //$$

*) Inverse of 5

$$26 = 5(5) + 1$$

$$1 = 26 - 5(5)$$

$$1 = 5(5) + 26(-1)$$

$$-5 + 26 = 21$$

$$5^{-1} \rightarrow \pmod{26}$$

$$= 21 //$$

* Inverse of 7

$$26 = 7(3) + 5$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$2 = 5 - 2(3 - 5)$$

$$1 = 5 - 2 \times 3 + 2 \times 5$$

$$1 = 3(5) - 2(3)$$

$$1 = 3(26 - 7(3)) - 2(3)$$

$$1 = 3 \times 26 - 3 \times 21 - 2(3)$$

$$1 = 3 \times 26 - 11 \times 7$$

$$1 = (-11) \times 7 + 3 \times 26$$

$$7^{-1} \rightarrow 15 \pmod{26}$$

$$15 //$$

* Inverse of 9

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 9 + 9(2) - 26$$

$$1 = 3(9) - 26$$

$$9^{-1} \rightarrow 3 \pmod{26}$$
$$= 3 //$$

*) Inverse of 11

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$3 = 11 - 4(2)$$

$$4 = 8(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$= 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3 \times 26 - 3 \times (11(2)) - 11$$

$$1 = 3 \times 26 - 7 \times 11$$

$$1 = (-7)11 + 3 \times 26$$

$$11^{-1} \rightarrow 19 \pmod{26}$$

19 //

*) Inverse 15

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 3(4) - 11$$

$$1 = 3(15 - 11(1)) - 11$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 3 \times 15 - 4(26 - 15)$$

$$1 = 7(15) - 4(26)$$

$$15^{-1} \rightarrow 7 \pmod{26}$$

7 //

* Inverse of 19

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 5 - 2 \times 7 + 2(5 \times 1)$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(19 - 2(7)) - 2(7)$$

$$1 = 3 \times 19 - 8(7)$$

$$1 = 3(19) - 8(26 - 19)$$

$$1 = 11(19) - 8(26)$$

$$19^{-1} \rightarrow 11 \pmod{26}$$

$$= 11 //$$

* Inverse of 21

$$26 = 21(1) + 5$$

$$21 = 5(4) + 1$$

$$1 = 21 - 5(4)$$

$$1 = 21 - 5(4)$$

$$1 = 21 - 4(26 - 21)$$

$$1 = 21 - 4(26) + 4(21)$$

$$1 = 5(21) - 4(26)$$

$$21^{-1} \rightarrow \pmod{26}$$

$$= 5 //$$

*) Inverse of 23

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2(1)$$

$$1 = 3 - (23 - 3(7))$$

$$1 = 8(3) - 23$$

$$1 = 8(26 - 23) - 23$$

$$1 = (-7)23 + 26(8)$$

$$26 - 7 = 19$$

$$23^{-1} \rightarrow 19 \pmod{26}$$

$$= 19 //$$

*) Inverse of 25

$$26 = 25(1) + 1$$

$$1 = 26 - 25$$

$$1 = (-1)25 + 26$$

$$26 - 1 = 25$$

$$25^{-1} \rightarrow 25 \pmod{26}$$

$$= 25 //$$

2) Plain = GOODLUCK.

$$P = (k_1 * P + k_2) \bmod 26$$

Plaintext

		$k_1 * P$	$+ k_2$	mod 26 mod 26
G	6	30	38	12
O	14	70	78	0
O	14	70	78	0
D	3	15	23	23
L	11	55	63	11
U	20	100	108	4
L	2	10	18	18
K	10	50	58	6

Ciphertext: M A A X L E S G

3) Affine cipher with $a=11$

$$P = H.I \quad b=5$$

$$k_1 = 11, k_2 = 5$$

$$C = E(k_1 * P + k_2) \bmod 26$$

Plain
text

		$k_1 * P$	$+ k_2$	mod 26	Cipher
H	7	77	82	4	E
I	8	88	93	15	P

== EP //

4) VCR encrypted using the affine function $9x + 2 \pmod{26}$

Find Plaintext:

$$P = K^{-1} (C - K_2) \pmod{26}$$

$$K_1 = 9, K_2 = 2$$

$$\begin{array}{l|l} 9^{-1} \pmod{26} & 1 = 9 - (26 - 9(2)) \\ 26 = 9(2) + 8 & 1 = 3(9) - 26 \\ 9 = 8(1) + 1 & \\ 1 = 9 - 8 & 9^{-1} \pmod{26} = 3 \end{array}$$

CP	Value	$K^{-1}(C - K_2)$	$\pmod{26}$	PT
V	20	$3(20 - 2) = 54$	2	C
C	2	$3(2 - 2) = 0$	0	A
R	7	$3(7 - 2) = 15$	15	T

$$PT = CAT //$$

5) Decrypt the given ciphertext
using Affine cipher $KD Y X J U W D E X Z Y$
and the key $3 \pmod{26}$ & 4

$$P = K^{-1} (C - K_2) \pmod{26}$$

$$CP = KD Y X J U W D E X Z Y$$

$$K_1 = 3 \quad K_2 = 4$$

$$3^{-1} \pmod{26}$$

$$26 = 3(8) + 2$$

$$2 = 26 - 3(8)$$

$$3 = 12(1) + 1$$

$$1 = 3 - 2$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3(9) - 26$$

$$3^{-1} \pmod{26} = 9$$

CP	Value	$K^{-1}(C - K_2)$	$\pmod{26}$	PT
K	10	54		
D	3	-9	2	E
Y	24	180	17	R
X	23	171	24	Y
J	9	45	15	P
U	20	144	19	T
W	22	162	4	O
D	3	-9	6	G
E	4	0	17	R
X	23	171	0	A
Z	25	189	15	P
Y	24	180	7	H
			24	Y

PT = CRYPTOGRAPHY

6) use $k_1 = 13$ (Multiplier $k_2 = 4$ Even).

Through it is not a valid key try to encrypt.

Encrypt the string "Input" & "alter" using the above key & find the what is the issue.

Encryption.

$$C = E(k_1 * P + k_2) \bmod 26$$

$$k_1 = 13$$

$$k_2 = 4$$

Plt	Value	$k_1 * P$	$+ k_2$	mod 26 Value.	CT
I	8	104	108	4	E
N	13	169	173	17	R
P	15	195	199	17	R
U	20	260	264	4	E
T	19	247	251	17	R
A	0	0	4	4	E
L	11	143	147	17	R
T	19	247	251	17	R
E	4	52	56	4	E
R	17	221	225	17	R

$k_1 = 13$ is not a prime with 26

The entire 26 letter alphabet is compressed into 2 letters (E & R)

4) Find the Inverse matrix for key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ mod 26

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3 \times 7) - (5 \times 2)$$

$$= 21 - 10$$

$$= 11$$

$$11^{-1} \text{ mod } 26$$

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 26 - 11(2) - (11) = -6(11) - (11) + 3 \times 26$$

$$1 = (-7)11 + 3 \times 26 \quad | \quad 7 + 26 = 19$$

$$11^{-1} \text{ mod } = 19$$

$$= 19/11$$

$$19 \begin{bmatrix} 3 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix whose are constants mod 26 $\begin{pmatrix} 11 & 13 \\ 2 & 3 \end{pmatrix}$

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$= (11 \times 3) - (13 \times 2)$$

$$= 33 - 26 = 7$$

$$7^{-1} \text{ mod } 26 \Rightarrow 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 3(26 - 7(3) - 1(7))$$

$$1 = (-11)7 + 3(26)$$

$$-11 + 26 = 15$$

$$7^{-1} \text{ mod } 26 = 15$$

$$15 \begin{bmatrix} 3 & -13 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 15 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

a) Given key is valid $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

$$= (2 \times 21) - (13 \times 14)$$

$$= 42 - 182$$

$$= -140$$

-140 is not co-prime with 26.

$$21 \times 2$$

$$42$$

$$13$$

$$14 \times 2$$

b) Encrypt using 2×2 Hill cipher for the Plaintext "SOLVED" using $\begin{pmatrix} 3 & 8 \\ 7 & 11 \end{pmatrix}$

Encrypt

$$C = P \times K \text{ mod } 26$$

$$= (\text{S})(\text{O})(\text{L})(\text{V})(\text{E})(\text{D})$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 11) - (7 \times 8)$$

$$= 33 - 56$$

$$= -23 \Rightarrow \text{co-prime with } 26$$

$$P = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 10 \\ 20 \end{bmatrix} = KU$$

2) $P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 21 \end{bmatrix} = \begin{bmatrix} 20 \\ 308 \end{bmatrix} \text{mod } 26.$$

$$= \begin{bmatrix} 19 \\ 22 \end{bmatrix} = TW$$

3) $P = \begin{bmatrix} F \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 10 \\ 9 \end{bmatrix} = KJ$$

Cipher text : KUTWKJ

3) ~~next~~

$$12 + 24$$

$$= 36 \quad 28 + 33 =$$

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25